

Machine Learning Basics

Machine Learning (ML) enables computers to learn from data and improve performance over time without being explicitly programmed.

Key Points

1. Supervised Learning: Learn from labeled data.
2. Unsupervised Learning: Discover patterns in unlabeled data.
3. Reinforcement Learning: Learn through interaction with environment.

Example Code

```
from sklearn.linear_model import LinearRegression
import numpy as np

X = np.array([[1], [2], [3], [4]])
y = np.array([2, 4, 6, 8])

model = LinearRegression()
model.fit(X, y)
print('Prediction:', model.predict([[5]]))
```

Conclusion

ML is transforming industries by automating decision-making and pattern recognition.